

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400968
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Weidinger; Gerald et al.

Component with dielectric layer for embedding in component carrier

Abstract

A component carrier includes a stack with at least one electrically insulating layer structure and/or at least one electrically conductive layer structure, and a component having one or more pads and at least one dielectric layer on at least one main surface of the component. The at least one dielectric layer does not extend beyond the main surface in a lateral direction. The dielectric layer at least partially covers one or more pads of the component. In addition, at least one electrically conductive contact extends through at least one opening in the dielectric layer up to at least one of the pads.

Inventors: Weidinger; Gerald (Leoben, AT), Zluc; Andreas (Leoben, AT)

Applicant: AT&S Austria Technologie & Systemtechnik Aktiengesellschaft (Leoben, AT)

Family ID: 1000008779768

Assignee: AT&SAustria Technologie & Systemtechnik Aktiengesellschaft (Leoben, AT)

Appl. No.: 18/357236

Filed: July 24, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230369235 A1	Nov. 16, 2023

Foreign Application Priority Data

EP	18193535	Sep. 10, 2018
----	----------	---------------

Related U.S. Application Data

continuation parent-doc US 17451012 20211015 US 11749613 child-doc US 18357236
continuation-in-part parent-doc US 16559399 20190903 US 11171092 20211109 child-doc US

Publication Classification

Int. Cl.: H01L23/538 (20060101); H01L21/48 (20060101); H01L21/56 (20060101); H01L23/00 (20060101); H01L23/31 (20060101); H01L23/373 (20060101)

U.S. Cl.:

CPC H01L23/5389 (20130101); H01L21/4853 (20130101); H01L21/565 (20130101); H01L21/568 (20130101); H01L23/3121 (20130101); H01L23/373 (20130101); H01L23/5386 (20130101); H01L24/19 (20130101); H01L24/20 (20130101); H01L2224/214 (20130101)

Field of Classification Search

CPC: H01L (23/3121); H01L (23/373); H01L (23/5386); H01L (24/19); H01L (21/4853); H01L (21/568); H01L (21/565); H01L (24/20); H01L (23/498-49894); H01L (23/538-5389); H01L (21/4846-4867); H01L (21/4807-481); H01L (23/16-26); H01L (2224/05001-05499)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6309912	12/2000	Chiou et al.	N/A	N/A
6354338	12/2001	Takemoto	N/A	N/A
8354338	12/2012	Hu et al.	N/A	N/A
2003/0013232	12/2002	Towle et al.	N/A	N/A
2005/0017347	12/2004	Morimoto et al.	N/A	N/A
2005/0028925	12/2004	Fernandes et al.	N/A	N/A
2006/0145328	12/2005	Hsu	N/A	N/A
2009/0267171	12/2008	Yean et al.	N/A	N/A
2010/0044845	12/2009	Funaya et al.	N/A	N/A
2012/0060908	12/2011	Crafts	N/A	N/A
2012/0313226	12/2011	Koizumi et al.	N/A	N/A
2013/0200534	12/2012	Shiobara	428/447	H01L 23/3135
2013/0337648	12/2012	Lin et al.	N/A	N/A
2014/0138758	12/2013	Uozaki	257/317	H10D 64/035
2014/0252641	12/2013	Lim	438/618	H01L 23/49822
2015/0008587	12/2014	Lin et al.	N/A	N/A
2016/0020110	12/2015	Lu	257/618	H01L 21/3086
2016/0155702	12/2015	Chen	N/A	N/A
2016/0262267	12/2015	Lin	N/A	H01L 24/00
2018/0040551	12/2017	Lin et al.	N/A	N/A
2018/0082988	12/2017	Cheng et al.	N/A	N/A
2018/0092220	12/2017	Wiedenhofer	N/A	H01L 24/20

2018/0315647	12/2017	Wang	N/A	H01L 23/53209
2019/0148301	12/2018	Huang et al.	N/A	N/A
2020/0098886	12/2019	Liu	N/A	H01L 29/785

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101299908	12/2007	CN	N/A
101364581	12/2008	CN	N/A
101998772	12/2010	CN	N/A
103515247	12/2013	CN	H01L 21/4857
105990159	12/2015	CN	N/A
106057768	12/2015	CN	H01L 21/4846
107295747	12/2016	CN	N/A
108364921	12/2017	CN	N/A
1240390	12/2004	TW	N/A
200618227	12/2005	TW	N/A
2015016344	12/2014	WO	N/A
2015026344	12/2014	WO	N/A
2016176512	12/2015	WO	N/A
2016176512	12/2015	WO	N/A

OTHER PUBLICATIONS

Kästner, M.; Communication in Application No. 18 193 535.4; pp. 1-5; Jun. 4, 2021; European Patent Office, Postbus 5818, 2280 HV Rijswijk, Netherlands. cited by applicant

Natanni, F.; Extended European Search Report in Application No. 18 193 535.4; pp. 1-9; Jan. 8, 2019, European Patent Office, 80298, Munich, Germany. cited by applicant

First Office Action in Application No. 201910851934.9; pp. 1-7; Mar. 6, 2023, China National Intellectual Property Administration; 6 Xitucheng Lu, Jimenqiao, Haidian District, Beijing 100088, China. cited by applicant

Office Action in Application No. 201910851934.9, pp. 1-6; Jan. 9, 2024, China National Intellectual Property Administration, No. 6, Xitucheng Lu, Jimenqiao Haidian District, Beijing City, 100088, China. cited by applicant

Notification for Patent Registration Formalities in Application No. 2019108519349, pp. 1-5; Jun. 25, 2024; China National Intellectual Property Administration; No. 6 Xitucheng Road, Haidian District, Beijing, 100080, P.R. China. cited by applicant

Translation of Notification for Patent Registration Formalities in Application No. 2019108519349, pp. 1-2; Jun. 25, 2024; China National Intellectual Property Administration; No. 6 Xitucheng Road, Haidian District, Beijing, 100080, P.R. China. cited by applicant

Primary Examiner: Ahmed; Shahed

Attorney, Agent or Firm: Smith Tempel Blaha LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of co-pending U.S. Non-provisional patent application Ser. No. 17/451,012 filed on Oct. 15, 2021, and U.S. Non-provisional patent application Ser. No. 16/559,399 filed on Sep. 3, 2019, and claims the benefit of and right of priority to European patent application no. 18 193 535.4, filed Sep. 10, 2018, which applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The invention relates to a method of manufacturing a component carrier and to a component carrier.

Technological Background

(2) In the context of growing product functionalities of component carriers equipped with one or more electronic components and increasing miniaturization of such components as well as a rising number of components to be mounted on or embedded in the component carriers such as printed circuit boards, increasingly more powerful array-like components or packages having several components are being employed, which have a plurality of contacts or connections, with ever smaller spacing between these contacts. Removal of heat generated by such components and the component carrier itself during operation becomes an increasing issue. At the same time, component carriers shall be mechanically robust and electrically reliable so as to be operable even under harsh conditions.

(3) In particular, efficiently embedding a component in a component carrier is an issue.

SUMMARY

(4) There may be a need to efficiently embed a component in a component carrier.

(5) According to an exemplary embodiment of the invention, a component carrier is provided which comprises a stack comprising at least one electrically insulating layer structure and/or at least one electrically conductive layer structure, a component (or multiple components) having one or more pads and at least one dielectric layer on at least one main surface of the component, wherein the at least one dielectric layer does not extend beyond the main surface in a lateral direction, wherein the dielectric layer at least partially covers the one or more pads of the component, and at least one electrically conductive contact extending through at least one opening in the dielectric layer up to at least one of the one or more pads.

(6) According to another exemplary embodiment of the invention, a method of manufacturing a component carrier is provided, wherein the method comprises forming a stack comprising at least one electrically conductive layer structure and/or at least one electrically insulating layer structure, embedding a component in the stack, wherein the component comprises at least one dielectric layer arranged on at least one main surface of the component and at least partially covering one or more pads of the component, and forming at least one opening in the dielectric layer and at least partially filling the at least one opening with at least one electrically conductive contact to thereby electrically connect at least one of the one or more pads of the component.

Overview of Embodiments

(7) In the context of the present application, the term “component carrier” may particularly denote any support structure which is capable of accommodating one or more components thereon and/or therein for providing mechanical support and/or electrical connectivity. In other words, a component carrier may be configured as a mechanical and/or electronic carrier for components. In particular, a component carrier may be one of a printed circuit board, an organic interposer, and an IC (integrated circuit) substrate. A component carrier may also be a hybrid board combining different ones of the above-mentioned types of component carriers.

(8) In the context of the present application, the term “dielectric layer” may particularly denote a layer covering at least part of a surface of a component and comprising or even consisting of an electrically insulating material. Such a dielectric layer may be applied as a coating or an attached film to the component and may cover a part of or even entire pads of the component via which one

or more pads the component can be electrically contacted. The dielectric layer may be a flat foil covering one or two opposing main surfaces of the component. Side walls of the component may or may not be covered by the dielectric layer. The dielectric layer may also be a closed shell fully circumferentially covering the component.

(9) In the context of the present application, the term “dielectric layer which does not extend beyond the main surface of a component in a lateral direction” may particularly denote that the dielectric layer only covers the surface of the component and does not extend significantly beyond lateral ends of the component. While the dielectric layer may optionally also cover a sidewall of the component connected to a main surface at an angle (in particular at a right angle), the dielectric layer of such an embodiment shall not laterally protrude significantly beyond such a sidewall, in particular not over a dimension being larger than a thickness of the dielectric layer.

(10) According to an exemplary embodiment of the invention, a component carrier is provided which has a component with one or more pads on a surface of the component which is covered with a dielectric layer being formed on the component already before embedding the component in a layer stack of the component carrier. By taking this measure, the component may be already prepared for proper electric contacting prior to the embedding procedure, so that after embedding the component in the (preferably laminated) layer stack of electrically conductive layer structures and/or electrically insulating layer structures, the obtained body does not necessarily be covered by a further electrically insulating layer structure in a separate lamination process prior to exposing the one or more pads of the component for contacting. By correspondingly configuring the dielectric layer (in particular by correspondingly adjusting material and/or thickness thereof), it may be thus possible to keep a number of lamination procedures for manufacturing the component carrier small. At the same time, also the entire thickness of the component carrier may be kept very compact. Also, the embedding height of the component can be precisely adjusted by correspondingly designing the mentioned dielectric layer. As a result, a simple manufacturability of the component carrier may be combined with a thin and compact shape thereof.

(11) In the following, further exemplary embodiments of the method and the component carrier will be explained.

(12) In an embodiment, a bottom surface of the dielectric layer is at the same vertical level and in alignment with a bottom surface of the stack. In particular when the component is located in a cavity of the stack, the dielectric layer thereof may be located entirely within the cavity so that its bottom surface flushes with a bottom surface of the stack. This provides a particularly compact architecture in vertical direction.

(13) In an embodiment, the dielectric layer comprises at least one of the group consisting of resin, in particular epoxy resin, a photo-imagable dielectric, and polyimide. Using a resin, in particular epoxy resin, as a material for the dielectric layer renders the coated or covered component particularly appropriate for component carrier technology, in particular PCB (printed circuit board) technology. The reason for this is that in PCB technology, such resin materials are also used as well, so that issues such as a thermal mismatch in terms of different values of the coefficient of thermal expansion of dielectric material of the component carrier, dielectric material bridges, etc. may be safely prevented. Moreover, such component carrier material can be properly processed by laser drilling, mechanically drilling, etc. as well as is compatible with copper filling. When a photo-imagable dielectric (such as a photoresist) is used as material for the dielectric layer, patterning the dielectric layer by a photoimaging process becomes possible. This simplifies the process of exposing the at least partially covered pad(s) of the component for establishing an electric contact with an environment. When the dielectric layer is made of a photo-imagable material, it may for instance comprise or consist of a photoresist. Also, the use of polyimide as material for the dielectric layer is advantageous, since also this material can be properly processed by laser or mechanically drilling and is compatible with copper plating technology.

(14) In an embodiment, the dielectric layer is made of a copper-plateable material. Hence, the

dielectric layer may be capable of serving as a support for plated copper deposited on the dielectric layer and/or in one or more openings extending therethrough.

(15) In an embodiment, the dielectric layer is made of a thermally conductive material, in particular having a value of the thermal conductivity of at least 1 W/mK. When the material of the dielectric layer is made of a highly thermally conductive material (in particular having a higher thermal conductivity than prepreg), the dielectric layer may also significantly contribute to the removal of heat generated by the component in an interior of the component carrier during operation. For instance, when the component is a semiconductor chip such as a processor, a considerable amount of heat may be generated in an interior of the component carrier and needs to be re-moved therefrom in order to prevent undesired effects such as overheating or thermal loads. Such a heat removal or heat spreading in the plate type component carrier may be promoted by a dielectric layer made of a properly thermally conductive material.

(16) In an embodiment, the dielectric layer is made of a laser drillable material. When the dielectric layer is made of a material which can be drilled by a laser, the openings may be easily and precisely manufactured by laser drilling. This further improves the compatibility of the component coated or covered with the dielectric layer with PCB technology.

(17) In an embodiment, a thickness of the dielectric layer is in a range between 0.5 μm and 100 μm , in particular in a range between 10 μm and 20 μm . On the one hand, the dielectric layer should not be too thin in order to prevent undesired uncoated regions of the component which might deteriorate the electric performance of the component carrier. On the other hand, the dielectric layer should not be too thick in order to keep the manufactured component carrier thin in a vertical direction. The mentioned ranges have turned out as a proper trade-off between these and other considerations.

(18) In an embodiment, the at least one electrically conductive contact comprises at least one of the group consisting of a via, in particular at least one of a laser via, a photo via and a plasma via, filled at least partially with electrically conductive material, and a metallic pillar, in particular a copper pillar. Thus, the external electric coupling of the embedded component may be accomplished by vias (in particular laser vias) formed and extending vertically through the dielectric layer.

Additionally or alternatively, it is also possible to use metal pillars, i.e. metallic posts, extending through the dielectric layer for contacting the pad(s) of the component.

(19) In an embodiment, the component has one or more pads under a respective dielectric layer on each of two opposing main surfaces of the component, wherein each respective dielectric layer at least partially covers the respective one or more pads on the respective main surface of the component. In such an embodiment, both opposing main surfaces of the component may be covered with a respective dielectric layer and may have pads on both of these two opposing main surfaces. This renders even sophisticated contact architectures possible. It is also possible that the entire circumferential surface of the component is covered with dielectric layer material, for instance by dipping the entire component in a liquid precursor for such a dielectric material. Alternatively, only the mentioned two opposing main surfaces may be covered partially or entirely with a respective dielectric layer, for instance by adhering sticky foils as dielectric layers to the main surfaces. In yet another embodiment, only one main surface of the component may be covered with the mentioned dielectric layer to keep the thick-ness of the component as thin as possible.

(20) In an embodiment, the method comprises forming the at least one electrically conductive contact without previously connecting at least one further electrically insulating layer structure to the at least one dielectric layer. Thus, it may be dispensable to attach one or more further electrically insulating layer structures to a bottom surface of both the stack and the embedded component before exposing the pads of the component with respect to an exterior electronic environment of the laminate-type component carrier. This renders the manufacturing process quick and easy and allows obtaining component carriers with very small vertical thickness.

(21) In an embodiment, the component is already provided with the at least one opening at the point of time of embedding the component. Therefore, the component with the one or more dielectric layers may already be provided with one or more openings which expose the pad(s) at the point of time of embedding the component in the stack. Thus, a later opening or exposing of the pads of the component by forming openings extending through the dielectric layer may become dispensable. Forming these access holes or openings may be easy when conditioning the still isolated or separate component.

(22) In an embodiment, the method comprises at least partially filling the at least one opening with electrically conductive material after embedding the component. For instance, this filling procedure may be accomplished by plating, in particular copper plating.

(23) In an embodiment, the method comprises forming the at least one opening by laser processing, in particular laser drilling. Laser drilling is a process which allows the formation of openings with high precision and in a short time.

(24) In an embodiment, the method comprises providing the stack with a cavity, closing at least part of a bottom of the cavity by a temporary carrier, and placing the component in the cavity so that at least one of the at least one dielectric layer is attached onto the temporary carrier. In such an embodiment, the cavity may be a through-hole extending through the entire stack (for instance a fully cured core). In order to temporarily fix the component in such a cavity, a sticky tape (with or without holes) may be attached to a lower main surface of the stack so as to partially or even entirely close the cavity. After that, the component may be attached to the temporary carrier.

(25) In an embodiment, the method comprises at least partially filling a gap in the cavity between the component and the stack with an additional filling medium, and thereafter removing the temporary carrier from the stack, the component and the filling medium. Such a filling medium may be a material which fixes the component in place within the cavity. After having applied and cured such a filling medium, the layer stack becomes stiffened and the temporary carrier is no longer needed for defining the position of the component in the cavity and can be removed. For instance, a sticky tape (as example for the temporary carrier) may be delaminated or peeled off from the rest of the component carrier being presently manufactured.

(26) In an embodiment, filling the gap is carried out by at least one of the group consisting of applying a liquid filling medium into the gap, and laminating an at least partially uncured electrically insulating layer structure to the stack and the component. Hence, the filling medium may for instance be liquid medium applied by a dispensing device or the like after having placed the component on the temporary carrier. For instance, such an adhesive may be an epoxy-based adhesive. Alternatively, a previously at least partially uncured electrically insulating layer structure (for instance a prepreg layer) may be laminated on both the stack and the component. During this lamination process, i.e. the application of heat and/or pressure, dielectric material of the previously at least partially uncured electrically insulating layer structure may melt or may become liquid and may also flow in the gap between component and stack. During this lamination, the previously at least partially uncured electrically insulating layer structure may be cured by cross-linking of a resin material thereof. After that, the cross-linked and cured material will re-solidify and may then fix the component in place in the cavity of the stack. It is also possible that the dielectric layer itself is made of an at least partially uncured material which may be cured during lamination so as to flow in the gaps and fill them while contributing to an adhesion between the constituents of the component carrier.

(27) In an embodiment, the method comprises embedding by placing the component between a flat support structure (in particular at least one of the layer structures of the stack or a temporary carrier) and at least one flat one of the layer structures. In such an embodiment, no cavity needs to be formed before embedding the component in the stack. In contrast to this, the present embodiment uses planar layer structures between which the component is embedded by lamination without the prior formation of a through-hole or a blind hole. No core, i.e. already fully cured

material, needs to be used in such an embodiment which may correspond to a coreless configuration.

(28) Still referring to the previously described embodiment, the embedding may comprise pressing the component into the at least one of the layer structures and/or into the support structure. Pressing the component into planar electrically insulating layer structures (for instance prepreg layers) above and/or below the component during lamination may become possible when the mentioned electrically insulating layer structures are not yet fully cured at the point of time of pressing the component into such layers. By the described embodiment in which the component is embedded by pressing it between two flat structures, formation of a cavity in terms of embedding becomes dispensable.

(29) In an embodiment, the at least one dielectric layer is in a fully cured state already prior to inserting the component in the stack. When the dielectric layer is already cured (for instance is FR4 material) at the point of time of embedding, it can be safely ensured that the dielectric layer stays in place and does not flow into gaps in a surrounding during the manufacturing process. This ensures a high spatial accuracy of the position of the component within the stack.

(30) In another embodiment, the at least one dielectric layer is in an at least partially uncured state when inserting the component in the stack. In such an alternative embodiment, the material of the dielectric layer may be cured by lamination, i.e. the application of heat and/or pressure, and may therefore contribute to the interior or intrinsic adhesion between the constituents of the component carrier. In such an embodiment, the pad(s) of the component may still be completely covered by the dielectric material when embedding the component into the stack.

(31) In an embodiment, the method comprises inserting the component in the stack in a condition in which the at least one dielectric layer is a continuous layer. The one or more openings extending through the dielectric layer and up to the pads may then be formed later, i.e. after the embedding procedure.

(32) Still referring to the previously described embodiment, the method comprises forming the at least one opening and the at least one electrically conductive contact after the inserting. This allows the dielectric layer to properly protect the contact surface of the component during the embedding procedure and to expose the contact surface only after the embedding.

(33) In an embodiment, at least partially filling the at least one opening with the at least one electrically conductive contact element comprises attaching a further electrically conductive layer structure (such as a copper foil) to a bottom of the stack and to at least one of the at least one dielectric layer, and at least partially plating (for instance galvanically plating) the at least one opening with electrically conductive material.

(34) In an embodiment, the method comprises providing the dielectric layer with an electrically insulating matrix (in particular a polymer removable by a laser beam) and an additive comprising a metal compound (in particular a metal compound being activatable by a laser beam), selectively treating a surface portion of the dielectric layer (in particular by irradiation with a laser beam along a predefined trajectory) to thereby locally remove material of the electrically insulating matrix while simultaneously locally activating the additive for promoting subsequent metal deposition, and selectively depositing metallic material on the locally activated additive. Descriptively speaking, the dielectric layer on the component may be treated by a laser beam in such a way that only in the regions of the laser treatment, a surface of the dielectric layer becomes laser activated. A subsequent deposition of metallic material (in particular copper) predominantly or even fully selectively occurs on the laser activated additive only, since the metallic compounds form seeds for such a metallization. For instance, such a metallization may be carried out in a current-less way (for instance in a copper bath or the like). When the additive of the metallic compound is electrically insulating, the manufactured metallization may form electrically conductive traces on the component carrier.

(35) In an embodiment, the method comprises connecting at least one further electrically insulating

layer structure and/or at least one further electrically conductive layer structure to at least one of a top side and a bottom side of the stack. Thus, a further build-up may be carried out after the embedding procedure in accordance with a desired application of the component carrier being manufactured. The mentioned connection of at least one further layer structure to one or both of the two opposing main surfaces of the stack with embedded component may be carried out symmetrically or asymmetrically.

(36) The material of the dielectric layer may also be functionalized. In other words, the material may be selected so that the dielectric layer fulfils at least one additional function. As an example, such a function may be a high frequency capability, a heat removal property, etc.

(37) The at least one component can be selected from a group consisting of an electrically non-conductive inlay, an electrically conductive inlay (such as a metal inlay, preferably comprising copper or aluminum), a heat transfer unit (for example a heat pipe), a light guiding element (for example an optical waveguide or a light conductor connection), an electronic component, or combinations thereof. For example, the component can be an active electronic component, a passive electronic component, an electronic chip, a storage device (for instance a DRAM or another data memory), a filter, an integrated circuit, a signal processing component, a power management component, an optoelectronic interface element, a voltage converter (for example a DC/DC converter or an AC/DC converter), a cryptographic component, a transmitter and/or receiver, an electromechanical transducer, a sensor, an actuator, a microelectromechanical system (MEMS), a microprocessor, a capacitor, a resistor, an inductance, a battery, a switch, a camera, an antenna, a logic chip, a light guide, and an energy harvesting unit. However, other components may be embedded in the component carrier. For example, a magnetic element can be used as a component. Such a magnetic element may be a permanent magnetic element (such as a ferromagnetic element, an antiferromagnetic element or a ferrimagnetic element, for instance a ferrite base structure) or may be a paramagnetic element. However, the component may also be a further component carrier, for example in a board-in-board configuration. One or more components may be surface mounted on the component carrier and/or may be embedded in an interior thereof. Moreover, also other than the mentioned components may be used as component.

(38) In an embodiment, the component carrier comprises a stack of at least one electrically insulating layer structure and at least one electrically conductive layer structure. For example, the component carrier may be a laminate of the mentioned electrically insulating layer structure(s) and electrically conductive layer structure(s), in particular formed by applying mechanical pressure, if desired supported by thermal energy. The mentioned stack may provide a plate-shaped component carrier capable of providing a large mounting surface for further components and being nevertheless very thin and compact. The term “layer structure” may particularly denote a continuous layer, a patterned layer or a plurality of non-consecutive islands within a common plane.

(39) In an embodiment, the component carrier is shaped as a plate. This contributes to the compact design, wherein the component carrier nevertheless provides a large basis for mounting components thereon. Furthermore, in particular a naked die as example for an embedded electronic component, can be conveniently embedded, thanks to its small thickness, into a thin plate such as a printed circuit board.

(40) In an embodiment, the component carrier is configured as one of the group consisting of a printed circuit board, and a substrate (in particular an IC substrate).

(41) In the context of the present application, the term “printed circuit board” (PCB) may particularly denote a component carrier (which may be plate-shaped (i.e. planar), three-dimensionally curved (for instance when manufactured using 3D printing) or which may have any other shape) which is formed by laminating several electrically conductive layer structures with several electrically insulating layer structures, for instance by applying pressure, if desired accompanied by the supply of thermal energy. As preferred materials for PCB technology, the

electrically conductive layer structures are made of copper, whereas the electrically insulating layer structures may comprise resin and/or glass fibers, so-called prepreg or FR4 material. The various electrically conductive layer structures may be connected to one another in a desired way by forming through-holes through the laminate, for instance by laser drilling or mechanical drilling, and by filling them with electrically conductive material (in particular copper), thereby forming vias as through-hole connections. Apart from one or more components which may be embedded in a printed circuit board, a printed circuit board is usually configured for accommodating one or more components on one or both opposing surfaces of the plate-shaped printed circuit board. They may be connected to the respective main surface by soldering. A dielectric part of a PCB may be composed of resin with reinforcing fibers (such as glass fibers).

(42) In the context of the present application, the term “substrate” may particularly denote a small component carrier having substantially the same size as a component (in particular an electronic component) to be mounted thereon. More specifically, a substrate can be understood as a carrier for electrical connections or electrical networks as well as component carrier comparable to a printed circuit board (PCB), however with a considerably higher density of laterally and/or vertically arranged connections. Lateral connections are for example conductive paths, whereas vertical connections may be for example drill holes. These lateral and/or vertical connections are arranged within the substrate and can be used to provide electrical and/or mechanical connections of housed components or unhoused components (such as bare dies), particularly of IC chips, with a printed circuit board or intermediate printed circuit board. Thus, the term “substrate” also includes “IC substrates”. A dielectric part of a substrate may be composed of resin with reinforcing spheres (such as glass spheres).

(43) In an embodiment, the at least one electrically insulating layer structure comprises at least one of the group consisting of resin (such as reinforced or non-reinforced resins, for instance epoxy resin or Bismaleimide-Triazine resin, more specifically FR-4 or FR-5), cyanate ester, polyphenylene derivate, glass (in particular glass fibers, multi-layer glass, glass-like materials), prepreg material, polyimide, polyamide, liquid crystal polymer (LCP), epoxy-based Build-Up Film, polytetrafluoroethylene (Teflon), a ceramic, and a metal oxide. Reinforcing materials such as webs, fibers or spheres, for example made of glass (multilayer glass) may be used as well. Although prepreg or FR4 are usually preferred, other materials may be used as well. For high frequency applications, high-frequency materials such as polytetrafluoroethylene, liquid crystal polymer and/or cyanate ester resins may be implemented in the component carrier as electrically insulating layer structure.

(44) In an embodiment, the at least one electrically conductive layer structure comprises at least one of the group consisting of copper, aluminum, nickel, silver, gold, palladium, and tungsten. Although copper is usually preferred, other materials or coated versions thereof are possible as well, in particular coated with supra-conductive material such as graphene.

(45) In an embodiment, the component carrier is a laminate-type body. In such an embodiment, the semifinished product or the component carrier is a compound of multiple layer structures which are stacked and connected together by applying a pressing force, if desired accompanied by heat.

(46) The aspects defined above and further aspects of the invention are apparent from the examples of embodiment to be described hereinafter and are explained with reference to these examples of embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1, FIG. 2, FIG. 3, and FIG. 4 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component, shown

in FIG. 5, according to an exemplary embodiment of the invention.

(2) FIG. 6 illustrates a cross-sectional view of a component carrier according to another exemplary embodiment of the invention.

(3) FIG. 7, FIG. 8, FIG. 9, FIG. 10 and FIG. 11 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component, shown in FIG. 12, according to another exemplary embodiment of the invention.

(4) FIG. 13, FIG. 14, FIG. 15, FIG. 16, FIG. 17, FIG. 18 and FIG. 19 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing component carriers with embedded component according to exemplary embodiments of the invention.

(5) FIG. 20, FIG. 21, FIG. 22, FIG. 23 and FIG. 24 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component according to another exemplary embodiment of the invention.

(6) FIG. 25, FIG. 26, FIG. 27, FIG. 28 and FIG. 29 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component according to still another exemplary embodiment of the invention.

(7) FIG. 30, FIG. 31, FIG. 32, FIG. 33, FIG. 34 and FIG. 35 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component according to another exemplary embodiment of the invention.

(8) FIG. 36, FIG. 37, FIG. 38, FIG. 39, FIG. 40, FIG. 41 and FIG. 42 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing component carriers with embedded component according to other exemplary embodiments of the invention.

(9) FIG. 43, FIG. 44 and FIG. 45 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component according to yet another exemplary embodiment of the invention.

(10) FIG. 46 and FIG. 47 illustrate plan views of structures obtained during carrying out a method of manufacturing a component carrier with an embedded component according to an exemplary embodiment of the invention.

DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS

(11) The illustrations in the drawings are schematically presented. In different drawings, similar or identical elements are provided with the same reference signs.

(12) Before, referring to the drawings, exemplary embodiments will be described in further detail, some basic considerations will be summarized based on which exemplary embodiments of the invention have been developed.

(13) According to an exemplary embodiment of the invention, an embedding architecture using dielectrically coated or covered components is provided.

(14) By using a component having a dielectric layer at the point of time of embedding this component in a component carrier stack, laminating procedures may be omitted in a build-up based on a temporary carrier. This may allow obtaining simply manufacturable and thin component carriers.

(15) In one embodiment, the following method of manufacturing a component carrier may be carried out.

(16) First, a stack comprising at least one electrically insulating layer structure and/or at least one electrically conductive layer structure (for instance a core) which has been provided with cavities in the form of through-holes or the like may be laminated with a temporary carrier. Thereafter, the components with the dielectric layer may be placed in these through-holes and on the temporary carrier, in particular with the dielectric layer being connected directly with the temporary carrier. The components may be adhered with the stack by an appropriate adhesive or resin filled in gaps between the stack and the component carrier. Thereafter, the temporary carrier may be removed. The surface may be metallized, for instance by carrying out a metal deposition procedure (for instance a chemical copper deposition procedure followed by a galvanic copper deposition

procedure). Additionally or alternatively, it is possible to laminate, for instance using prepreg foils, copper foils and/or RCC (Resin Coated Copper) foils. Thereafter, a patterning, contacting and further component carrier manufacturing procedure may be carried out.

(17) In another embodiment, the following method may be carried out: A stack composed of at least one electrically insulating layer structure and/or at least one electrically conductive layer structure (such as a fully cured core) may be laminated on a temporary carrier after having forming through-holes or the like extending through the stack as cavities. The components with the dielectric layers may be placed with the dielectric layer facing the temporary carrier on the bottom. A lamination with at least one further layer structure (for instance a prepreg sheet, a resin sheet with a copper foil or an RCC foil) may be performed. Thereafter, the temporary carrier may be removed. For instance, an uppermost copper layer may be removed, for example by etching. Thereafter, a main surface may be metallized using a metal deposition procedure (for instance a chemical copper deposition procedure followed by a galvanic copper deposition procedure). After that, patterning, contact formation and continuation of the component carrier manufacturing procedure may be carried out.

(18) It should be mentioned that by providing the component with a dielectric layer on one main surface of two opposing main surfaces of the component, pads of the component may be contacted directly after embedding, without the need to carry out an additional dielectric lamination procedure beforehand. This allows obtaining particularly thin component carriers. In an embodiment, a specific coating (for instance using a palladium complex) may be used, and the stack (for instance a core) may be roughened before metallization (for instance by a plasma process).

(19) The above-described embodiments refer to an embedding of the component with dielectric layer using cavities formed in a stack. However, other exemplary embodiments of the invention may embed the component with dielectric layer without the use of cavities. In such embodiments, it is possible to use in particular one or more of the following materials: Resin sheets, asymmetric prepregs, RCC (Resin Coated Copper) materials, Sumitomo materials, TD002 prepreg, coatings (i.e. liquid resin compounds), mold materials (for instance on the basis of resin mixtures), etc. In an embedding procedure without cavities in a stack, the components with the one or more dielectric layers may be pressed into adjacent material (for instance of planar layers) during lamination. For contacting the embedded components, it is for instance possible to form copper-filled laser vias, copper-filled plasma vias and/or copper pillars.

(20) In yet another exemplary embodiment, it is possible to embed pre-patterned components. In such an embodiment, components having a dielectric layer may be embedded after patterning of the dielectric layer. For instance, such a patterning can be performed by a photo or plasma process forming one or more openings in the dielectric layers for exposing the pad(s) of the component. Contacting may be carried out by laser drilling with a subsequent copper filling procedure.

(21) In still another exemplary embodiment of the invention, it is possible to embed components using photovias. In such an embodiment, it is for instance possible to employ a photo-imagable dielectric layer (for instance made of a photoresist) in which the vias exposing the pads of the component can be formed by imaging and stripping. Filling the vias may be carried out during a subsequent copper procedure.

(22) In yet another exemplary embodiment of the invention, embedding of components may be accomplished using plasma vias. The vias for exposing the pads of the component may be formed by applying a mask followed by a plasma etching procedure. Filling the vias may be carried out during a copper process.

(23) In still another exemplary embodiment of the invention, embedding of components may be accomplished using copper pillars extending through openings of the dielectric layer. Vias for contacting the pads of a component may be realized by forming a dielectric layer on the component which is already provided with copper pillars.

(24) In yet another exemplary embodiment of the invention, embedding of components may be accomplished using a laser patternable dielectric layer. For such an embodiment, the components may be provided with a polymeric dielectric layer which is doped with a (preferably electrically non-conductive) laser activatable metal compound as additive to the polymer. At a position where a laser beam impinges on such a plastic, the plastic matrix can be disintegrated into volatile reaction products in a surface region. At the same time, metal seeds may be split off from the additives which are present in a micro-rough surface. These metal particles form a seed for a subsequent metallization. In a current-less copper bath, the partial surfaces treated by the laser processing may be used for forming electrically conductive traces. A corresponding patterning procedure may be embodied as Laser Direct Patterning process.

(25) In still another exemplary embodiment of the invention, a thermally conductive coating may be used as material for the dielectric layer. When the dielectric layer is equipped with or made of thermally highly conductive particles such as AlN, Al₂O₃, BN, the heat removal properties of the components may be improved.

(26) In an embodiment, it is possible to form the dielectric layer covering at least a part of a surface of the component using a material in a B-stage configuration. In other words, the dielectric material of the dielectric layer may still be in an at least partially uncured state, for instance may be provided as a not yet fully cross-linked epoxy resin. The dielectric layer may then contribute to the intra-stack adhesion of the component carrier being manufactured.

(27) FIG. 1 to FIG. 5 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102**, shown in FIG. 5, according to an exemplary embodiment of the invention.

(28) Referring to FIG. 1, a stack **104** composed of electrically conductive layer structures **108** and an electrically insulating layer structure **106** is shown. In the shown embodiment, the stack **104** may be a core of fully cured resin (optionally comprising reinforcing particles such as glass fibers) constituting the electrically insulating layer structure **106** which is covered on both opposing main surfaces with a respective copper foil constituting a respective one of the electrically conductive layer structures **108**. As can be taken from FIG. 1, a cavity **118** is formed as a through hole in the stack **104**.

(29) Furthermore, a component **102** (such as a semiconductor chip) is shown which is to be embedded in the cavity **118** formed in the stack **104**. The component **102** comprises a dielectric layer **112** covering only the entire lower main surface of the component **102**. For example, a thickness “d” of the dielectric layer **112** may be 10 μm. The dielectric layer **112** extends over the entire main surface but does not extend beyond the main surface in a lateral direction corresponding to a horizontal direction according to FIG. 1. The dielectric layer **112** therefore also covers pads **110** formed on a lower main surface of the component **102**. The dielectric layer **112** may be in a fully cured state already prior to inserting the component **102** in the stack **104**. For instance, the dielectric layer **112** may be made of a fully cured resin (such as an epoxy resin), optionally comprising reinforcing particles such as glass fibers. In such an embodiment, the material of the dielectric layer **112** is prevented from flowing away during a lamination procedure which ensures that the component **102** remains precisely in place during lamination. Alternatively, the dielectric layer **112** may be in an at least partially uncured state when inserting the component **102** in the stack **104**. In such an embodiment, the material of the dielectric layer **112** may contribute to the adhesion of the constituents of the component carrier **100**.

(30) Moreover, a temporary carrier **120** (here embodied as a sticky tape) is shown which has been attached to a lower main surface of the stack **104** so as to close the entire bottom of the cavity **118**.

(31) As can be taken from an arrow **160** in FIG. 1, the component **102** with the dielectric layer **112** on its bottom surface is to be placed in the cavity **118**.

(32) Referring to FIG. 2, the component **102** is now placed in the cavity **118** so that the dielectric layer **112** is attached onto the temporary carrier **120**. The component **102** is hence accommodated

in the stack **104** in a condition in which the dielectric layer **112** is a continuous layer arranged exclusively but completely on exactly one main surface of the component **102**. As a result, the dielectric layer is attached to a sticky surface of the temporary carrier **120**, for instance a sticky tape. As can be taken from FIG. 2, a bottom surface of the dielectric layer **112** is at the same vertical level and in alignment with a bottom surface of the stack **104**.

(33) Referring to FIG. 3, a gap **122** in the cavity **118** between the component **102** and the stack **104** is filled with an additional filling medium **161**. This procedure of filling the gap **122** may be carried out by applying a liquid filling medium into the gap **122** and by curing the liquid filling medium. The gaps **122** are thus filled with adhesive material which is cured for fixing the component **102** in place in the cavity **118**.

(34) Referring to FIG. 4, the temporary carrier **120** is then removed from the stack **104**, the embedded component **102** and the cured filling medium **161**. The adhesive tape forming the temporary carrier **120** is hence removed by peeling it off.

(35) Referring to FIG. 5, openings **116** (not shown in FIG. 5, compare however for instance FIG. 23) are formed in the dielectric layer **112**, for instance by laser drilling. Subsequently, the openings **116** are filled with electrically conductive material to thereby form electrically conductive contacts **114** for electrically connecting the pads **110** of the component **102** with an electronic periphery of the component carrier **100** being presently formed. The openings **116** in the dielectric layer **112** are thus filled with electrically conductive material after embedding the component **102**. Hence, the dielectric layer **112** is made of a copper-plateable and laser drillable material. Advantageously, the electrically conductive contacts **114** can be formed without previously connecting a further electrically insulating layer structure to the dielectric layer **112** and the stack **104**.

(36) As can be taken from FIG. 5 as well, it is possible to attach a respective further electrically conductive layer structure **108** (such as a further copper foil) to both a top and a bottom of the stack **104** and to the dielectric layer **112** and the component **102**, respectively. Thereafter, the openings **116** may be plated with electrically conductive material such as copper. FIG. 5 shows the result of a metallization and contacting procedure and hence the component carrier **100**.

(37) Referring to FIG. 6, a component carrier **100** according to another embodiment is shown in which the component **102** has pads **110** and a respective dielectric layer **112** on each of two opposing main surfaces of the component **102**. Each respective dielectric layer **112** partially covers the respective pads **110** on the respective main surface of the component **102**. FIG. 6 hence shows an alternative component carrier **100** which differs from the embodiment of FIG. 5 in that electrically conductive contacts **114** extending through openings **116** in the respective dielectric layer **112** are formed on both opposing main surfaces of the component **102**.

(38) Optionally and although not shown in FIG. 1 to FIG. 6, it is possible to laminate at least one further electrically insulating layer structure **106** (such as at least one prepreg layer) and/or at least one further electrically conductive layer structure **108** (such as at least one further copper foil) to the top side and/or the bottom side of the component carrier **100** shown in FIG. 6.

(39) FIG. 7 to FIG. 12 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102**, shown in FIG. 12, according to another exemplary embodiment of the invention.

(40) The procedure shown in FIG. 7 and FIG. 8 correspond to the procedures shown in FIG. 1 and FIG. 2, respectively.

(41) Referring to FIG. 9, a further electrically insulating layer structure **164** (such as a prepreg sheet) is laminated on the upper main surface of the structure shown in FIG. 9. Due to the curing of the resin material of the further electrically insulating layer structure **164** (by the application of pressure and/or heat) the material of the further electrically insulating layer structure **164** re-melts and becomes flowable so as to fill the gaps **122** until it re-solidifies after completion of a cross-linking process.

(42) According to FIG. 10, temporary carrier **120** is now removed since the component **102** is now

fixed in place due to the resin filling of the gaps **122** as a result of the lamination process described referring to FIG. **9**.

(43) The structure of FIG. **11** is then obtained by a metallization and contacting procedure.

(44) The component carrier **100** according to FIG. **12** has electrically conductive contacts **166** formed in an upper portion of the component carrier **100**. These electrically conductive contacts **166** are formed at laterally adjacent portions of the component **102**, i.e. extending through the further electrically insulating layer structure **164** connected by the procedure described above referring to FIG. **9**.

(45) FIG. **13** to FIG. **19** illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing component carriers **100** with embedded component **102** according to other exemplary embodiments of the invention. FIG. **13** to FIG. **19** show structures obtained during manufacturing a component carrier **100** by an embedding procedure using a copper covered core as stack **104**. Fixing a component **102** in place may be accomplished by filling gaps **122** of a cavity **118** with liquid adhesive or by laminating a previously at least partially uncured electrically insulating layer structure.

(46) According to FIG. **13**, a stack **104** is shown which is configured as a core covered with patterned copper layers on both opposing surfaces thereof.

(47) As can be taken from FIG. **14**, the core with the cavity **118** is attached to temporary carrier **120**. Component **102** with dielectric layer **112** on a lower main surface thereof is placed in the cavity **118** and is attached face down (i.e. with pads **110** oriented downwardly) to the sticky tape forming the temporary carrier **120**. According to FIG. **14**, the further build-up is accomplished by lamination of a further electrically insulating layer structure **164**, such as a prepreg sheet, and a further electrically conductive layer structure **168**, for instance a copper foil.

(48) In contrast to this, the further build-up established according to FIG. **15** is carried out by lamination of only a resin or prepreg sheet as further electrically insulating layer structure **164**.

(49) FIG. **16** shows the result of the lamination procedure according to FIG. **14**, whereas FIG. **17** shows the result of the lamination procedure according to FIG. **15**.

(50) FIG. **18** shows a component carrier **100** (or a preform thereof) obtained by removing the temporary carrier **120** from the structure shown in FIG. **16**.

(51) FIG. **19** shows a component carrier **100** (or a preform thereof) according to another exemplary embodiment of the invention obtained by removing the temporary carrier **120** from the structure shown in FIG. **17**.

(52) FIG. **20** to FIG. **24** illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to still another exemplary embodiment of the invention.

(53) FIG. **20** shows, see reference numeral **170**, how liquid adhesive **161** is applied in gaps **122** between the component **102** and the stack **104**.

(54) As can be taken from FIG. **21**, such liquid adhesive **161** may be applied not only in the gaps **122**, but also underfill voids at a lower side of the component **102** as well as covers the component **102** on an upper side.

(55) FIG. **22** shows the result of the application of the liquid adhesive **161** after curing.

Furthermore, the temporary carrier **120** has meanwhile been removed from the lower main surface of the build-up shown in FIG. **21**.

(56) As can be taken from a detail shown in FIG. **23**, laser vias have now been formed in the lower main surface of the shown layer structure, wherein these laser vias form frustoconical openings **116** extending from an exterior main surface of the shown structure up to the previously covered pads **110** of the component **102**. By taking this measure, the pads **110** are partially exposed.

(57) The component carrier **100** according to the detail illustrated in FIG. **24** can be obtained by filling the openings **116** with an electrically conductive material such as copper, thereby forming the electrically conductive contacts **114**. This may be accomplished by a plating process.

(58) FIG. 25 to FIG. 29 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to another exemplary embodiment of the invention.

(59) Referring to FIG. 25, the component **102** is already provided with the openings **116** at the point of time of embedding the component **102**. As shown in FIG. 25, the dielectric layer **112** of the component **102** is inserted in a cavity **118** of a stack **104** and is attached on a sticky surface of a temporary carrier **120** closing the cavity **118** on a bottom side. The dielectric layer **112** on the component **102** is already foreseen with through-holes or openings **116** extending up to the pads **110** of the component **102**.

(60) FIG. 26 shows the result of the described pick-and-place assembly.

(61) In order to obtain the structure shown in FIG. 27, a further electrically insulating layer structure **164** is laminated to an upper main surface of the structure shown in FIG. 26. Thereby, also the gaps **122** are filled with resin material or the like.

(62) The structure shown in FIG. 28 may then be obtained by removing the temporary carrier **120**. The pads **110** are thereby exposed towards the electronic environment, since the openings **116** are now exposed. The openings **116**, which may be photovias, may therefore allow an access to the component **102** without the need of attaching a further electrically insulating layer structure to the lower main surface of FIG. 28.

(63) The component carrier **100** shown in FIG. 29 can be obtained by plating electrically conductive material such as copper on a lower main surface of the structure shown in FIG. 28. As a result, the openings **116** are filled with copper material or the like, thereby forming electrically conductive contacts **114**. If desired, the electrically conductive layer structures **108** forming the upper and lower main surfaces, respectively, of the component carrier **100** according to FIG. 29 may be patterned.

(64) FIG. 30 to FIG. 35 illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to an exemplary embodiment of the invention. Referring to FIG. 30 to FIG. 35, a method of manufacturing component carrier **100** according to yet another exemplary embodiment of the invention will be described in the following, which is based on the use of a component **102** which is not placed in a cavity **118** but which is sandwiched between two planar layers.

(65) Referring to FIG. 30, a bottom of the component **102** is placed on a flat main surface of temporary carrier **120**. The component **102** is embedded between the layer structures **106**, **108** on top and the temporary carrier **120** on bottom. More specifically, the embedding comprises pressing the component **102** into the layer structure **106** (which may be a resin or prepreg sheet). As shown in FIG. 30, the component **102** with the already applied dielectric layer **112** is hence placed between temporary carrier **120** on the lower side and electrically insulating layer structure **106** as well as electrically conductive layer structure **108** (such as a copper foil) on an upper side. The electrically insulating layer structure **106** may be an at least partially uncured layer structure, for instance a prepreg layer.

(66) In order to obtain the layer structure shown in FIG. 31, the constituents according to FIG. 30 may be connected by lamination, in particular the application of pressure and/or heat. The integral body shown in FIG. 31 is thereby obtained. During this process, the component **102** with the dielectric layer **112** on the lower main surface thereof is pressed into the electrically insulating layer structure **106** and is thereby embedded.

(67) In order to obtain the structure shown in FIG. 32, the temporary carrier **120** may be removed from a lower main surface of the structure shown in FIG. 31.

(68) FIG. 33 shows a detail of the structure shown in FIG. 32.

(69) As can be taken from FIG. 34, the structure shown in FIG. 33 may then be made subject to a patterning procedure for forming the openings **116** extending through the dielectric layer **112** up to the pads **110**, for instance by laser processing. By taking this measure, the pads **110** of the

component **102** are exposed.

(70) In order to obtain the component carrier **100** according to FIG. **35**, the lower main surface of the structure shown in FIG. **34** is made subject to a metal deposition procedure. Thereby, the openings **116** are filled with metallic material, preferably copper, thereby forming electrically conductive contacts **114**. By such a plating procedure, also the lower main surface of the stack **104** may be covered with electrically conductive material.

(71) For the coreless processing according to FIG. **30** to FIG. **35**, the component **102** may be provided with a vertical thickness “D” of preferably 10 μm to 50 μm, see FIG. **30**.

(72) FIG. **36** to FIG. **42** illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to another exemplary embodiment of the invention. Referring to FIG. **36** to FIG. **42**, a method of manufacturing a component carrier **100** according to yet another exemplary embodiment of the invention will be described in which the components **102** with dielectric layer **112** are embedded without forming of a cavity **118**. This manufacturing process may be carried out by implementing photovias.

(73) FIG. **36**, FIG. **37** and FIG. **38** corresponds to the procedures described above referring to FIG. **30**, FIG. **31**, and FIG. **32**, respectively.

(74) Removing material from the upper main surface of the structure shown in FIG. **38** allows obtaining the component carrier **100** as shown in FIG. **39**.

(75) FIG. **40** shows a detailed view of a portion of the structure of FIG. **38**.

(76) FIG. **41** shows a detail of the structure shown in FIG. **39**.

(77) In order to obtain the component carrier **100** shown in FIG. **42**, a (for instance copper) plating procedure can be carried out. The plated electrically conductive material fills the openings **116** to thereby form electrically conductive contacts **114**. Also, the upper and the lower main surface of the component carrier **100** of FIG. **42** is covered with electrically conductive material such as copper as a result of the described plating procedure.

(78) FIG. **43** to FIG. **45** illustrate cross-sectional views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to an exemplary embodiment of the invention. In this embodiment, copper pillars are used as electrically conductive contacts **114**. From the beginning of the described process onwards, the component **102** with the dielectric layer **112** is already provided with the copper pillars as electrically conductive contacts **114** extending through the dielectric layer **112**. In other words, the copper pillars already form part of the components **102** at the point of time before embedding the component **102** in stack **104**.

(79) Referring to FIG. **43**, the described component **102** is placed between a temporary carrier **120** located on a bottom of the component **102** and an arrangement of layer structures **106**, **108** located above the component **102**. The layer structures **106**, **108** are composed of an uncured electrically insulating layer structure **106** (for instance a prepreg layer) and an electrically conductive layer structure **108** (such as a copper foil).

(80) The structure in FIG. **44** can be obtained by connecting the constituents shown in FIG. **43** by lamination followed by a removal of the temporary carrier **120**.

(81) FIG. **45** shows the result of a plating or a further lamination procedure applied to the structure of FIG. **44**. As a result, both opposing main surfaces of the structure shown in FIG. **44** are covered by electrically conductive material such as copper. It is possible that the structure **100** according to FIG. **45** is further processed, for instance patterned.

(82) FIG. **46** and FIG. **47** illustrate plan views of structures obtained during carrying out a method of manufacturing a component carrier **100** with an embedded component **102** according to an exemplary embodiment of the invention.

(83) FIG. **46** shows a plan view of a component **102** with a dielectric layer **112** and openings **116** for contacting pads **110** (not shown in FIG. **46**). In the shown embodiment, the dielectric layer **112**

is provided as a doped material with an electrically insulating matrix and an additive comprising a metal compound in the matrix. Such a component **102** may be used with both a coreless manufacturing process as well as with a manufacturing process using a core.

(84) After an embedding procedure of the component **102** shown in FIG. **46** in a stack **104**, and now referring to FIG. **47**, a surface portion or trajectory of the dielectric layer **112** may be selectively processed by a laser beam (not shown) to thereby locally remove material of the electrically insulating matrix while simultaneously locally activating the additive for promoting subsequent metal deposition. By this activation, it becomes possible to subsequently selectively deposit metallic material (such as copper) on the locally activated additive only. As a result, electrically conductive traces **182** may be formed for establishing a desired electric contact task. In the shown embodiment, the electrically conductive traces **182** are also electrically coupled with the electrically conductive contacts **114** in the previous openings **116** for establishing an electric contact with the pads **110**.

(85) It is also shown in FIG. **47** that also an electrically insulating layer structure **106** of the stack **104** may be provided as a doped material with an electrically insulating matrix and an additive comprising a metal compound in the matrix. By taking this measure, the electrically conductive traces **182** may be formed partially on the dielectric layer **112**, and partially on the electrically insulating layer structure **106**.

(86) In contrast to conventional approaches in which an electrically insulating layer structure on a bottom surface of a component is applied by lamination after embedding, an exemplary embodiment of the invention employs a component with dielectric layer applied to the component already at a point of time of the embedding. This allows manufacturing very thin laminate type component carriers with embedded components. The manufacturing process is significantly simplified. Such a manufacturing architecture may be used for all kind of component carriers, in particular of PCB type, with embedded components in which a very thin component carrier and a simple manufacturing procedure are desired.

(87) It should be noted that the term “comprising” does not exclude other elements or steps and the article “a” or “an” does not exclude a plurality. Also, elements described in association with different embodiments may be combined.

(88) Implementation of the invention is not limited to the preferred embodiments shown in the figures and described above. Instead, a multiplicity of variants is possible which use the solutions shown and the principle according to the invention even in the case of fundamentally different embodiments.

Claims

1. A component carrier, comprising: a stack comprising at least one electrically insulating layer structure and at least one electrically conductive layer structure; a component having one or more pads and at least one dielectric layer on at least one main surface of the component, wherein the dielectric layer covers the one or more pads of the component; at least one electrically conductive contact extending through at least one opening in the dielectric layer up to at least one of the one or more pads; and a filling medium filling a cavity in the at least one electrically insulating layer structure surrounding the component and the dielectric layer, wherein a sidewall of the at least one electrically insulating layer structure is opposed to a sidewall of the dielectric layer with the filling medium present between the sidewall of the at least one electrically insulating layer structure and the sidewall of the dielectric layer; wherein the filling medium is in contact with a sidewall of the component; and wherein the filling medium is not in contact with the one or more pads of the component and the at least one main surface of the component, where the at least one dielectric layer is arranged.
2. The component carrier according to claim 1, comprising at least one of the following features:

wherein a bottom surface of the dielectric layer is at the same vertical level and in alignment with a bottom surface of the stack; wherein the dielectric layer comprises at least one of the group consisting of resin, epoxy resin, a photo-imagable dielectric, and polyimide; wherein the dielectric layer is made of a copper-plateable material; wherein the dielectric layer is made of a thermally conductive material having a value of the thermal conductivity of at least 1 W/m K; wherein the dielectric layer is made of a laser drillable material; wherein a thickness of the dielectric layer is in a range between 0.5 μm and 100 μm ; wherein the at least one electrically conductive contact comprises at least one of the group consisting of a via, filled at least partially with electrically conductive material, and a metallic pillar; wherein the component has one or more pads under a respective dielectric layer on each of two opposing main surfaces of the component.

3. The component carrier according to claim 1, comprising at least one of the following features: the at least one electrically insulating layer structure comprises at least one of the group consisting of reinforced or non-reinforced resin, epoxy resin or Bismaleimide-Triazine resin, FR-4, FR-5, cyanate ester, polyphenylene derivate, glass, prepreg material, polyimide, polyamide, liquid crystal polymer, epoxy-based Build-Up Film, polytetrafluoroethylene, a ceramic, and a metal oxide; wherein the component is selected from a group consisting of an electronic component, an electrically non-conductive and/or electrically conductive inlay, a heat transfer unit, an energy harvesting unit, an active electronic component, a passive electronic component, an electronic chip, a storage device, a filter, an integrated circuit, a signal processing component, a power management component, an optoelectronic interface element, a voltage converter, a cryptographic component, a transmitter and/or receiver, an electromechanical transducer, an actuator, a microelectromechanical system, a microprocessor, a capacitor, a resistor, an inductance, an accumulator, a switch, a camera, an antenna, a magnetic element, a light guiding element, a further component carrier and a logic chip; the component carrier is shaped as a plate; the component carrier is configured as a printed circuit board, or a substrate.

4. The component carrier according to claim 1, further comprising: at least one electrically conductive layer structure comprising at least one of the group consisting of copper, aluminum, nickel, silver, gold, palladium, and tungsten; wherein the filling medium is in contact with the at least one electrically conductive layer structure.

5. The component carrier according to claim 1, wherein the at least one dielectric layer does not extend beyond the main surface of the component in a lateral direction.

6. The component carrier according to claim 1, wherein the sidewall of the component and the sidewall of the dielectric layer are aligned to each other.

7. A method of manufacturing a component carrier, the method comprising: providing a stack having at least one electrically insulating layer structure with a cavity and at least one electrically conductive layer structure on a temporary carrier; embedding a component in the cavity, the component having at least one dielectric layer arranged on at least one main surface of the component and covering one or more pads of the component, the at least one dielectric layer arranged with an opening such that at least one electrically conductive contact extends through the opening to at least one of the pads; and filling the cavity between the component and the at least one electrically insulating layer structure with a filling medium, wherein a sidewall of the at least one electrically insulating layer structure is opposed to a sidewall of the dielectric layer so that the filling medium is present between the sidewall of the at least one electrically insulating layer structure and the sidewall of the dielectric layer, wherein the filling medium is in contact with a sidewall of the component; wherein the filling medium is not in contact with the one or more pads of the component and the at least one main surface of the component, where the at least one dielectric layer is arranged; and removing the temporary carrier.

8. The method according to claim 7, comprising at least one of the following features: wherein the component is provided with at least one opening at the point of time of embedding the component, at least partially filling the at least one opening with electrically conductive material after

embedding the component; forming the at least one opening by laser processing.

9. The method according to claim 7, wherein filling the cavity is carried out by applying a liquid filling medium into the cavity and curing the liquid filling medium.

10. The method according to claim 9, wherein during embedding the component in the cavity, the component is placed in the cavity so that the at least one dielectric layer is attached onto the temporary carrier.

11. The method according to claim 7, wherein filling the cavity is carried out by laminating an at least partially uncured electrically insulating layer structure to the at least one electrically insulating layer structure and the component.

12. The method according to claim 7, comprising at least one of the following features: wherein the at least one dielectric layer is in a fully cured state already prior to embedding the component; wherein the at least one dielectric layer is in an at least partially uncured state when embedding the component.

13. The method according to claim 7, further comprising: embedding the component in a condition in which the at least one dielectric layer is a continuous layer.

14. The method according to claim 13, further comprising: forming at least one opening in the dielectric layer and at least partially filling the at least one opening with at least one electrically conductive contact forming the at least one opening and the at least one electrically conductive contact after the embedding.

15. The method according to claim 7, further comprising: connecting at least one further electrically conductive layer structure to at least one of a top side and a bottom side of the component carrier.

16. The method according to claim 7, further comprising: providing the dielectric layer with an electrically insulating matrix and an additive comprising a metal compound; selectively treating a surface portion of the dielectric layer to thereby locally remove material of the electrically insulating matrix while simultaneously locally activating the additive for promoting subsequent metal deposition; and selectively depositing metallic material on the locally activated additive.
